

Quiz 3

CSC354 · Machine Learning

TOTAL MARKS
100

DURATION
30 min

DIFFICULTY
MEDIUM

PASSING THRESHOLD
75%

STUDENT NAME

STUDENT ID

SECTION

DATE

INSTRUCTIONS

- All questions are compulsory.
- Total time: 30 minutes.
- Total marks: 100.

Questions

CLO-5 Develop a reasonable size project using appropriate machine learning technique.

1. Develop a Python program to implement a recommendation system using collaborative filtering on a movie rating dataset. The system should be able to predict the ratings for movies that users have not yet rated based on their past behavior and similar user behaviors. Your solution must include data preprocessing, model training with appropriate hyperparameters tuning, and evaluation of the model's performance using mean absolute error (MAE) metric.
2. Develop a Python program to implement a recommendation system using collaborative filtering on a movie rating dataset. The system should be able to predict ratings for movies that users have not yet rated based on the preferences of similar users. Your implementation should include data preprocessing, model training with Singular Value Decomposition (SVD), and evaluation metrics such as Mean Absolute Error (MAE) or Root Mean Squared Error (RMSE).